

CIL MT 2026 — FULL-LENGTH MOCK CBT 3

Coal India Limited | Management Trainee | System Discipline (Post Code 14) | Advt. No. 03/2026

Total questions: 200 | **Total marks:** 200 | **Total time:** 3 hours (180 minutes), single sitting

Paper-I — General Knowledge/Awareness, Reasoning, Numerical Ability, General English	Q1 – Q100 (100 marks)
Paper-II — Professional Knowledge, System (CS/IT)	Q101 – Q200 (100 marks)

Instructions — follow these exactly to make the practice count

1. Set a single timer for 180 minutes and do not pause it. Both papers are attempted in one sitting, exactly as in the CBT.
 2. Each question carries 1 mark. **There is no negative marking** — leave nothing blank. Mark your best guess on anything you cannot solve.
 3. Both papers must be qualified **separately**: 40 marks for UR/EWS, 35 for OBC-NCL, 30 for SC/ST/PwD, out of 100 in each paper. Score each paper on its own before looking at the total.
 4. Suggested split: 75 minutes on Paper-I, 95 minutes on Paper-II, 10 minutes to sweep unattempted questions. Adjust after your first attempt.
 5. The answer key is at the end of this document. Do not open it until the timer stops.
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PAPER-I — GENERAL APTITUDE (Q1 - Q100)

SECTION A — GENERAL KNOWLEDGE / AWARENESS

- Q1.** The Neyveli lignite mines are located in
(A) Kerala (B) Tamil Nadu (C) Karnataka (D) Andhra Pradesh
- Q2.** The Simon Commission arrived in India in the year
(A) 1928 (B) 1935 (C) 1930 (D) 1919
- Q3.** Coal is popularly referred to as
(A) white gold (B) green fuel (C) black diamond (D) blue fuel
- Q4.** Non-coking coal in India is graded on the basis of
(A) moisture content only (B) Gross Calorific Value (C) ash content only (D) sulphur content
- Q5.** The Singrauli coalfield is located mainly in
(A) Odisha (B) Maharashtra (C) Madhya Pradesh (D) Jharkhand
- Q6.** Coal is formed by
(A) the compression of animal remains only (B) the evaporation of sea water (C) volcanic activity (D) the carbonisation of buried plant remains under heat and pressure
- Q7.** The Directorate General of Mines Safety (DGMS) functions under which Union Ministry?
(A) Ministry of Coal (B) Ministry of Mines (C) Ministry of Home Affairs (D) Ministry of Labour and Employment
- Q8.** Deficiency of iodine in the diet causes
(A) anaemia (B) scurvy (C) rickets (D) goitre
- Q9.** The first Governor-General of independent India was
(A) C. Rajagopalachari (B) Lord Wavell (C) Dr. Rajendra Prasad (D) Lord Mountbatten
- Q10.** Which subsidiary of Coal India Limited has its headquarters at Bilaspur?
(A) CCL (B) SECL (C) MCL (D) WCL
- Q11.** The Goods and Services Tax (GST) came into effect in India from
(A) 1 April 2016 (B) 1 July 2017 (C) 1 April 2017 (D) 1 January 2018
- Q12.** The headquarters of Mahanadi Coalfields Limited (MCL) is at
(A) Sambalpur (B) Cuttack (C) Bhubaneswar (D) Talcher
- Q13.** Coal India Limited functions under which Union Ministry?
(A) Ministry of Mines (B) Ministry of Heavy Industries (C) Ministry of Coal (D) Ministry of Power
- Q14.** The Dandi March (Salt Satyagraha) was launched in the year
(A) 1920 (B) 1928 (C) 1942 (D) 1930
- Q15.** The headquarters of Coal India Limited is located at
(A) Ranchi (B) New Delhi (C) Dhanbad (D) Kolkata
- Q16.** Central Coalfields Limited (CCL) has its headquarters at
(A) Bokaro (B) Patna (C) Dhanbad (D) Ranchi
- Q17.** The Coal Mines Provident Fund Organisation is headquartered at
(A) Kolkata (B) Dhanbad (C) New Delhi (D) Nagpur
- Q18.** Shares of Coal India Limited were first listed on Indian stock exchanges in
(A) 2005 (B) 2013 (C) 2010 (D) 2008
- Q19.** The place receiving the highest average annual rainfall in India is
(A) Mawsynram (B) Agumbe (C) Pasighat (D) Cherrapunji
- Q20.** The highest civilian award of India is the
(A) Padma Vibhushan (B) Padma Bhushan (C) Param Vir Chakra (D) Bharat Ratna
- Q21.** Coal India Limited was incorporated in the year
(A) 1971 (B) 1980 (C) 1973 (D) 1975
- Q22.** Abolition of untouchability is provided for under which Article of the Constitution?
(A) Article 17 (B) Article 19 (C) Article 16 (D) Article 15
- Q23.** The National Mineral Development Corporation (NMDC) is engaged primarily in the mining of
(A) limestone (B) coal (C) iron ore (D) bauxite
- Q24.** The Quit India Movement was launched in
(A) 1942 (B) 1930 (C) 1945 (D) 1935
- Q25.** Fundamental Duties were inserted into the Constitution by which amendment?
(A) 52nd Amendment (B) 44th Amendment (C) 24th Amendment (D) 42nd Amendment

SECTION B — REASONING

- Q26.** A person facing south turns left, then turns left again. He is now facing
(A) east (B) south (C) west (D) north
- Q27.** Statements: All squares are rectangles. All rectangles are polygons. Which conclusion follows?
(A) Some rectangles are not polygons (B) All polygons are squares (C) No square is a polygon (D) All squares are polygons

- Q28.** If FRIEND is coded as HTKGPF, then CANDLE is coded as
(A) ECPFNG (B) DBOEMF (C) EDRFNG (D) ECPFMG
- Q29.** Pointing to a photograph, a man said, "She is the daughter of my grandfather's only son." How is she related to the man?
(A) Cousin (B) Sister (C) Aunt (D) Niece
- Q30.** Find the next term: 2, 6, 12, 20, 30, ?
(A) 42 (B) 44 (C) 36 (D) 40
- Q31.** Statements: All engineers are graduates. Some graduates are teachers. Which conclusion follows?
(A) No engineer is a teacher (B) Some engineers are teachers (C) All graduates are engineers (D) None of these follows
- Q32.** Statements: No cat is a dog. All dogs are animals. Which conclusion follows?
(A) All animals are dogs (B) No animal is a cat (C) Some animals are not cats (D) Some cats are dogs
- Q33.** Choose the odd one out.
(A) Hockey (B) Chess (C) Football (D) Cricket
- Q34.** Find the next term: 1, 4, 9, 16, 25, ?
(A) 30 (B) 49 (C) 35 (D) 36
- Q35.** If 'P × Q' means P is the mother of Q, and 'P + Q' means P is the brother of Q, then in 'M × N + O', how is M related to O?
(A) Grandmother (B) Mother (C) Sister (D) Aunt
- Q36.** Choose the odd one out.
(A) Copper (B) Zinc (C) Brass (D) Iron
- Q37.** A man walks 6 km south and then 8 km west. His distance from the starting point is
(A) 2 km (B) 10 km (C) 14 km (D) 12 km
- Q38.** In a row of 25 persons, A is 11th from the left end and B is 8th from the right end. How many persons are seated between A and B?
(A) 5 (B) 7 (C) 8 (D) 6
- Q39.** If $5 * 3 = 16$ and $7 * 4 = 22$, then $9 * 5 = ?$
(A) 28 (B) 26 (C) 32 (D) 30
- Q40.** B is the wife of A and C is the son of A. How is C related to B?
(A) Son (B) Nephew (C) Brother (D) Father
- Q41.** Ram's father's sister's husband is Ram's
(A) cousin (B) brother (C) uncle (D) grandfather
- Q42.** Pointing to a woman, a boy said, "She is the mother of my father's only son." How is the woman related to the boy?
(A) Mother (B) Aunt (C) Sister (D) Grandmother
- Q43.** On a standard die, opposite faces total 7. If 2 is on the top face, the bottom face shows
(A) 3 (B) 4 (C) 6 (D) 5
- Q44.** P is 5th from the left end of a row and Q is 3rd to the right of P. What is Q's position from the left end?
(A) 8th (B) 6th (C) 7th (D) 9th
- Q45.** Pen : Write :: Knife : ?
(A) Kitchen (B) Steel (C) Cut (D) Sharp
- Q46.** Your mother's brother's son is your
(A) nephew (B) cousin (C) brother-in-law (D) uncle
- Q47.** Statements: Some books are pens. All pens are red. Which conclusion follows?
(A) Some books are red (B) All red things are pens (C) No book is red (D) All books are red
- Q48.** Find the next term: 120, 99, 80, 63, ?
(A) 50 (B) 46 (C) 48 (D) 44
- Q49.** Find the next term: 100, 96, 88, 76, ?
(A) 56 (B) 64 (C) 60 (D) 58
- Q50.** Find the next term: 4, 9, 19, 39, ?
(A) 79 (B) 80 (C) 78 (D) 59

SECTION C — NUMERICAL ABILITY

- Q51.** The LCM of 12, 15 and 20 is
(A) 40 (B) 180 (C) 60 (D) 120
- Q52.** The average of 4, 8, 12, 16 and 20 is
(A) 15 (B) 14 (C) 12 (D) 10
- Q53.** A speed of 90 km/h is equivalent to
(A) 30 m/s (B) 20 m/s (C) 15 m/s (D) 25 m/s
- Q54.** An article is sold for ₹850 after a discount of 15% on the marked price. The marked price is
(A) ₹1,000 (B) ₹977.50 (C) ₹1,050 (D) ₹950
- Q55.** A shopkeeper allows a discount of 20% on the marked price and still earns a profit of 20%. If the cost price is ₹400, the marked price is
(A) ₹600 (B) ₹625 (C) ₹576 (D) ₹540

- Q56.** 5% of 5% of 400 is
(A) 10 (B) 0.1 (C) 20 (D) 1
- Q57.** 40% of 250 is
(A) 90 (B) 110 (C) 125 (D) 100
- Q58.** An article is sold for ₹1,200 at a profit of 20%. The cost price is
(A) ₹1,000 (B) ₹1,100 (C) ₹960 (D) ₹1,020
- Q59.** 15% of 400 + 25% of 200 equals
(A) 110 (B) 130 (C) 120 (D) 100
- Q60.** Pipe A can fill a tank in 10 hours and pipe B in 15 hours. Opened together, they fill it in
(A) 8 hours (B) 6 hours (C) 5 hours (D) 12.5 hours
- Q61.** The value of $45 \div 0.9$ is
(A) 5 (B) 4.5 (C) 40.5 (D) 50
- Q62.** A population of 1,000 grows at 10% per annum. After 2 years it becomes
(A) 1,100 (B) 1,210 (C) 1,220 (D) 1,200
- Q63.** The average of the first 10 natural numbers is
(A) 6.5 (B) 6 (C) 5 (D) 5.5
- Q64.** An article marked at ₹500 is sold after successive discounts of 10% and 20%. The selling price is
(A) ₹375 (B) ₹360 (C) ₹400 (D) ₹350
- Q65.** A sum of money doubles itself in 8 years under simple interest. The rate of interest per annum is
(A) 15% (B) 10% (C) 12% (D) 12.5%
- Q66.** Pipe A can fill a tank in 6 hours while pipe B can empty it in 12 hours. With both open, the tank is filled in
(A) 18 hours (B) 6 hours (C) 12 hours (D) 4 hours
- Q67.** The value of 0.5×0.02 is
(A) 1.0 (B) 0.001 (C) 0.1 (D) 0.01
- Q68.** A car covers 300 km in 5 hours. Its average speed is
(A) 55 km/h (B) 50 km/h (C) 60 km/h (D) 65 km/h
- Q69.** The area of a triangle with base 10 cm and height 6 cm is
(A) 30 cm² (B) 60 cm² (C) 16 cm² (D) 32 cm²
- Q70.** The square root of 1764 is
(A) 42 (B) 44 (C) 38 (D) 46
- Q71.** 20% of 250 is
(A) 60 (B) 50 (C) 40 (D) 45
- Q72.** A fair die is rolled once. The probability of getting a prime number is
(A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$
- Q73.** Two cars start simultaneously from two points 200 km apart and move towards each other at 60 km/h and 40 km/h. They meet after
(A) 2.5 hours (B) 4 hours (C) 1 hour (D) 2 hours
- Q74.** The simple interest on ₹3,000 at 10% per annum amounts to ₹900. The time period is
(A) 2.5 years (B) 2 years (C) 3 years (D) 4 years
- Q75.** The length of the diagonal of a square of side 10 cm is
(A) $5\sqrt{2}$ cm (B) 10 cm (C) 20 cm (D) $10\sqrt{2}$ cm

SECTION D — GENERAL ENGLISH

Directions (Q76-Q77): Read the following passage and answer the questions that follow.

A public undertaking that trains its recruits for a year before confirming them is making a calculated investment. The cost is borne up front, while the return depends entirely on retention. Service bonds exist not to punish departure but to make the arithmetic work.

- Q76.** According to the passage, service bonds are intended primarily to
(A) penalise employees (B) increase salaries (C) shorten the training period (D) protect the employer's training investment
- Q77.** The phrase "borne up front" means the cost is
(A) recovered immediately (B) shared equally (C) avoided altogether (D) incurred at the beginning

Directions (Q78-Q79): Read the following passage and answer the questions that follow.

Safety training in mines is often treated as paperwork to be completed rather than a habit to be formed. Yet accident records show that the majority of incidents follow lapses in routine procedure, not rare equipment failure. A drill that is rehearsed until it becomes instinctive saves more lives than a manual that is merely signed.

- Q78.** According to the passage, most mine accidents are caused by
(A) lapses in routine procedure (B) rare equipment failures (C) shortage of trained staff (D) inadequate manuals
- Q79.** The tone of the passage towards existing safety training is
(A) critical (B) approving (C) indifferent (D) humorous

Directions (Q80-Q81): Read the following passage and answer the questions that follow.

Mechanisation in opencast mining has raised output per worker sharply over the last two decades. It has also changed the nature of the workforce required, replacing large numbers of manual labourers with smaller teams of skilled operators and technicians. The gain in productivity is therefore inseparable from a question about employment.

Q80. The passage suggests that mechanisation has

(A) reduced total coal output (B) had no effect on employment (C) increased manual labour requirements (D) altered the skill profile of the workforce

Q81. The word “inseparable” in the passage implies that the two issues are

(A) unrelated (B) closely linked (C) easily solved (D) recently discovered

Q82. Government by the people is called

(A) oligarchy (B) autocracy (C) monarchy (D) democracy

Q83. A medicine that counteracts the effect of a poison is called an

(A) anaesthetic (B) antibiotic (C) antiseptic (D) antidote

Q84. Identify the part containing an error: “The list of items / were sent / to the manager. / No error”

(A) were sent (B) to the manager (C) The list of items (D) No error

Q85. Choose the word most opposite in meaning to VOLUNTARY.

(A) Compulsory (B) Charitable (C) Willing (D) Optional

Q86. Choose the word most opposite in meaning to GENEROUS.

(A) Humble (B) Wealthy (C) Miserly (D) Kind

Q87. The idiom “a blessing in disguise” means

(A) an undeserved reward (B) a hidden threat (C) something good that at first seemed bad (D) a religious ceremony

Q88. Choose the correct passive form of: “They are building a new bridge.”

(A) A new bridge is being built by them (B) A new bridge is built by them (C) A new bridge has been building by them (D) A new bridge was being built by them

Q89. The idiom “once in a blue moon” means

(A) at night only (B) very frequently (C) unexpectedly (D) very rarely

Q90. Fill in the blank: “The meeting was postponed ___ next Monday.”

(A) till (B) since (C) at (D) in

Q91. Choose the word most similar in meaning to DILIGENT.

(A) Talkative (B) Cheerful (C) Careless (D) Hardworking

Q92. A person who talks too much is said to be

(A) garrulous (B) taciturn (C) fastidious (D) meticulous

Q93. Choose the word most similar in meaning to MEAGRE.

(A) Heavy (B) Generous (C) Scanty (D) Plentiful

Q94. Choose the correctly spelt word.

(A) Occurence (B) Occurance (C) Occurrence (D) Occurrence

Q95. The idiom “at the eleventh hour” means

(A) at the last possible moment (B) exactly on time (C) late at night (D) very early

Q96. Fill in the blank: “Neither of the two candidates ___ present at the interview.”

(A) were (B) are (C) have been (D) was

Q97. The idiom “to beat about the bush” means

(A) to avoid coming to the main point (B) to work in a garden (C) to attack someone verbally (D) to search thoroughly

Q98. Choose the word most similar in meaning to CANDID.

(A) Secretive (B) Rude (C) Doubtful (D) Frank

Q99. The idiom “to burn the midnight oil” means

(A) to waste resources (B) to sleep excessively (C) to cause a fire (D) to work late into the night

Q100. Choose the word most opposite in meaning to FRUGAL.

(A) Extravagant (B) Thrifty (C) Careful (D) Modest

SECTION 1 — ENGINEERING MATHEMATICS

- Q101.** The number of ways of selecting 3 objects out of 8 distinct objects is
(A) 112 (B) 336 (C) 24 (D) 56
- Q102.** The eigenvalues of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are
(A) 1 and 3 (B) 1 and 2 (C) 0 and 4 (D) 2 and 2
- Q103.** The total number of binary relations that can be defined on a set of 3 elements is
(A) 512 (B) 9 (C) 64 (D) 27
- Q104.** The number of edges in the complete graph K_7 is
(A) 14 (B) 21 (C) 42 (D) 49
- Q105.** A fair coin is tossed 4 times. The probability of getting exactly 2 heads is
(A) $1/2$ (B) $5/16$ (C) $1/4$ (D) $3/8$
- Q106.** The relation “congruent modulo 5” defined on the set of integers is
(A) a partial order only (B) antisymmetric but not reflexive (C) not transitive (D) an equivalence relation
- Q107.** The number of ways in which 5 distinct books can be arranged on a shelf is
(A) 120 (B) 720 (C) 25 (D) 60
- Q108.** The number of edges in the complete bipartite graph $K_{3,3}$ is
(A) 9 (B) 12 (C) 18 (D) 6
- Q109.** For the system $AX = B$ where A is square with $\det(A) = 0$ and $B \neq 0$, the system has
(A) a unique solution (B) always infinitely many solutions (C) exactly two solutions (D) either no solution or infinitely many solutions
- Q110.** The number of onto (surjective) functions from a set of 4 elements to a set of 3 elements is
(A) 36 (B) 64 (C) 24 (D) 81
- Q111.** For a binomial distribution with $n = 10$ and $p = 0.4$, the mean is
(A) 2.4 (B) 10 (C) 0.4 (D) 4
- Q112.** Two fair dice are rolled. The probability that the sum of the numbers is 7 is
(A) $1/12$ (B) $1/9$ (C) $5/36$ (D) $1/6$

SECTION 2 — DIGITAL LOGIC

- Q113.** The number of possible minterms for a Boolean function of 4 variables is
(A) 16 (B) 4 (C) 8 (D) 32
- Q114.** The number of output lines in a 3-to-8 decoder is
(A) 3 (B) 16 (C) 8 (D) 6
- Q115.** The SUM output of a half adder with inputs A and B is
(A) $A \oplus B$ (B) $A \cdot B$ (C) $A + B$ (D) $(A \cdot B)'$
- Q116.** The 2's complement of the 4-bit number 0110 is
(A) 0110 (B) 1001 (C) 1010 (D) 1101
- Q117.** The minimum number of flip-flops required to design a MOD-12 counter is
(A) 4 (B) 12 (C) 3 (D) 6
- Q118.** The race-around condition occurs in a
(A) level-triggered JK flip-flop with $J = K = 1$ (B) SR latch (C) T flip-flop with $T = 0$ (D) D flip-flop
- Q119.** The minimum number of 2-input NAND gates required to implement a 2-input XOR function is
(A) 3 (B) 4 (C) 6 (D) 5
- Q120.** The number of select (control) lines required for a 4-to-1 multiplexer is
(A) 2 (B) 3 (C) 4 (D) 1

SECTION 3 — COMPUTER ORGANIZATION & ARCHITECTURE

- Q121.** A processor supports 64 distinct instructions. The minimum number of bits required for the opcode field is
(A) 7 (B) 8 (C) 6 (D) 5
- Q122.** Memory interleaving is used mainly to improve
(A) cache size (B) memory reliability (C) effective memory bandwidth (D) memory capacity
- Q123.** During a DMA transfer, the system bus is controlled by
(A) the cache controller (B) the interrupt controller (C) the DMA controller (D) the CPU
- Q124.** The addressing mode most suitable for accessing successive elements of an array inside a loop is
(A) Immediate (B) Register (C) Direct (D) Indexed
- Q125.** Under ideal conditions, the speedup obtained from a k -stage instruction pipeline approaches
(A) k^2 (B) $\log k$ (C) $2k$ (D) k
- Q126.** Booth's algorithm is used for
(A) normalization of mantissa (B) multiplication of signed binary numbers (C) floating point addition (D) binary division

- Q127.** A direct-mapped cache of size 32 KB uses a block size of 32 bytes. The number of index bits required is
(A) 15 (B) 10 (C) 5 (D) 8
- Q128.** The throughput of a linear instruction pipeline is determined by
(A) the number of stages alone (B) the slowest stage (C) the fastest stage (D) the average stage delay
- Q129.** The number of address lines required to address 1 MB of byte-addressable memory is
(A) 20 (B) 24 (C) 16 (D) 18
- Q130.** In a write-back cache
(A) main memory is updated only when the block is replaced (B) every write updates main memory immediately (C) a dirty bit is not required (D) writes bypass the cache entirely
- Q131.** Cache memory is normally implemented using
(A) magnetic storage (B) SRAM (C) DRAM (D) EEPROM
- Q132.** Which of the following is a characteristic feature of RISC architecture?
(A) Complex microprogrammed control for most instructions (B) A very large number of addressing modes (C) Fixed-length instructions (D) Variable-length instructions
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SECTION 4 — PROGRAMMING & DATA STRUCTURES

- Q133.** The worst-case height of a binary search tree containing n nodes is
(A) $\log_2 n$ (B) $\sqrt{n}$ (C) $n - 1$ (D) $n/2$
- Q134.** Elements 1, 2, 3 are pushed onto a stack in that order, with pops interleaved arbitrarily. Which output sequence is NOT possible?
(A) 3, 2, 1 (B) 3, 1, 2 (C) 1, 3, 2 (D) 2, 3, 1
- Q135.** In a full binary tree in which every internal node has exactly 2 children, if there are 10 leaf nodes then the total number of nodes is
(A) 20 (B) 18 (C) 21 (D) 19
- Q136.** In a C program, memory obtained through `malloc()` is allocated from the
(A) heap (B) code segment (C) data segment (D) stack
- Q137.** The in-order traversal of a binary search tree yields the keys in
(A) sorted ascending order (B) random order (C) level order (D) reverse sorted order
- Q138.** On a machine where pointers are 64 bits wide, the value of `sizeof(int *)` in C is
(A) 8 (B) equal to `sizeof(int)` (C) 2 (D) 4
- Q139.** A recursive function written without a base case will most likely cause
(A) a compile-time error (B) heap fragmentation only (C) a run-time stack overflow (D) an infinite loop using no extra memory
- Q140.** The level-order traversal of a binary tree is implemented using
(A) a BST (B) a max-heap (C) a stack (D) a queue
- Q141.** A binary tree with n nodes has how many NULL child pointers?
(A) $n + 1$ (B) $n - 1$ (C) n (D) $2n$
- Q142.** The maximum number of nodes in a binary tree of height h (root at height 0) is
(A) $2h + 1$ (B) 2^h (C) $2^h - 1$ (D) $2^{(h+1)} - 1$
- Q143.** In C, the value returned by `strlen("hello")` is
(A) 4 (B) 5 (C) 6 (D) undefined
- Q144.** The postfix form of the expression $(A + B) * (C - D)$ is
(A) $+AB-CD*$ (B) $AB+CD-*$ (C) $ABCD+-*$ (D) $AB+C-D*$
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SECTION 5 — ALGORITHMS

- Q145.** Quicksort is applied to an already sorted array of n elements with the first element chosen as pivot. The running time is
(A) $O(n)$ (B) $O(\log n)$ (C) $O(n \log n)$ (D) $O(n^2)$
- Q146.** The maximum number of swaps performed by selection sort on n elements is
(A) $n - 1$ (B) n^2 (C) $n^2/2$ (D) $n \log n$
- Q147.** Which of the following sorting algorithms is stable?
(A) Heapsort (B) Selection sort (C) Quicksort (D) Merge sort
- Q148.** Kruskal's algorithm, implemented with sorting and union-find, runs in time
(A) $O(V^2)$ (B) $O(V^3)$ (C) $O(E \log E)$ (D) $O(E + V)$
- Q149.** The solution of $T(n) = 2T(n/2) + n$, $T(1) = 1$ is
(A) $\Theta(n)$ (B) $\Theta(\log n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2)$
- Q150.** Merging two sorted arrays of sizes m and n requires, in the worst case,
(A) $m + n - 1$ comparisons (B) mn comparisons (C) $\max(m, n)$ comparisons (D) $m + n$ comparisons
- Q151.** The solution of the recurrence $T(n) = 4T(n/2) + n^2$ is
(A) $\Theta(n^2)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2 \log n)$ (D) $\Theta(n^3)$
- Q152.** The maximum number of comparisons made by binary search on a sorted array of 1000 elements is
(A) 999 (B) 100 (C) 500 (D) 10

Q153. Prim's algorithm implemented with an adjacency matrix and no priority queue runs in
(A) $O(V + E)$ (B) $O(V^2)$ (C) $O(V^3)$ (D) $O(E \log V)$

Q154. The best-case time complexity of insertion sort is
(A) $O(n^2)$ (B) $O(n)$ (C) $O(n \log n)$ (D) $O(1)$

Q155. The solution of the recurrence $T(n) = T(n/2) + 1$ is
(A) $\Theta(\log n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n)$ (D) $\Theta(1)$

Q156. The worst-case number of comparisons in a linear search over n elements is
(A) n (B) $n/2$ (C) $n \log n$ (D) $\log n$

SECTION 6 — THEORY OF COMPUTATION

Q157. Regular languages are closed under
(A) intersection only (B) complementation only (C) union, intersection and complementation (D) union only

Q158. In the Chomsky hierarchy, a Type-3 grammar generates
(A) regular languages (B) context-free languages (C) context-sensitive languages (D) recursively enumerable languages

Q159. Deterministic pushdown automata accept
(A) all context-sensitive languages (B) all context-free languages (C) a proper subset of the context-free languages (D) only regular languages

Q160. The language $L = \{a^n b^n \mid n \geq 1\}$ is
(A) finite (B) context-free but not regular (C) regular (D) not context-free

Q161. A language accepted by a Turing machine that halts on every input is
(A) recursive (decidable) (B) necessarily context-free (C) recursively enumerable but not recursive (D) not recursively enumerable

Q162. Which class of languages is NOT closed under complementation?
(A) Finite languages (B) Recursive languages (C) Context-free languages (D) Regular languages

Q163. The grammar $S \rightarrow aSb \mid \epsilon$ generates the language
(A) $\{a^n b^m \mid n, m \geq 0\}$ (B) $(ab)^*$ (C) $\{a^n b^n \mid n \geq 0\}$ (D) all strings over $\{a, b\}$

Q164. In the worst case, the number of states in a DFA equivalent to an n -state NFA can be as large as
(A) n^2 (B) $n!$ (C) n (D) 2^n

SECTION 7 — COMPILER DESIGN

Q165. Left factoring of a grammar is performed to make it suitable for
(A) LR parsing (B) predictive (LL(1)) parsing (C) lexical analysis (D) peephole optimization

Q166. The compiler phase that constructs the parse tree is
(A) semantic analysis (B) code optimization (C) syntax analysis (D) lexical analysis

Q167. A recursive descent parser is an example of a
(A) bottom-up parser (B) top-down parser (C) LR parser (D) operator precedence parser

Q168. Elimination of left recursion from a grammar is essential for
(A) code optimization (B) LR parsing (C) lexical analysis (D) top-down parsing

Q169. Constant folding is an optimization performed
(A) by the operating system (B) at run time only (C) during linking (D) at compile time

Q170. An activation record created on the run-time stack typically contains
(A) only global variables (B) local variables, parameters and the return address (C) generated machine code (D) the complete symbol table

Q171. An L-attributed syntax-directed definition permits inherited attributes to be computed from
(A) the parent and left siblings (B) no other nodes (C) right siblings only (D) children only

Q172. An LR parser uses which data structure to hold states and symbols?
(A) Heap (B) Queue (C) Stack (D) Hash table

SECTION 8 — OPERATING SYSTEM

Q173. As the time quantum in Round Robin scheduling is made very small,
(A) context switch overhead increases (B) it behaves like SJF (C) starvation of long processes increases (D) context switch overhead decreases

Q174. Internal fragmentation is an inherent problem of
(A) pure segmentation (B) paging (C) both equally (D) neither

Q175. A process executes the statements `fork(); fork();`. The total number of processes in existence (including the original) is
(A) 3 (B) 4 (C) 8 (D) 2

Q176. The working set model is used primarily to
(A) speed up CPU scheduling (B) improve cache hit ratio (C) allocate disk blocks (D) prevent thrashing

Q177. The standard semaphore solution to the bounded-buffer producer-consumer problem uses how many semaphores?
(A) 1 (B) 2 (C) 4 (D) 3

- Q178.** The Banker's algorithm is used for
(A) deadlock detection (B) deadlock avoidance (C) page replacement (D) deadlock recovery
- Q179.** A transition of a process from the running state to the ready state is caused by
(A) process termination (B) an I/O request (C) a page fault (D) expiry of the time quantum
- Q180.** Which of the following is NOT an inter-process communication mechanism?
(A) Paging (B) Pipes (C) Shared memory (D) Message passing
- Q181.** Contiguous allocation of disk files provides
(A) no fragmentation at all (B) mandatory use of an index block (C) slow direct access (D) fast direct access but suffers from external fragmentation
- Q182.** In a system where every resource has exactly one instance, deadlock is detected by
(A) finding a cycle in the wait-for graph (B) checking the ready queue length (C) running the Banker's algorithm (D) counting page faults
- Q183.** The operating system module that performs the actual switching of the CPU to the selected process is the
(A) loader (B) long-term scheduler (C) linker (D) dispatcher
- Q184.** Mutual exclusion implemented using the TestAndSet instruction is an example of
(A) a compiler directive (B) a purely software solution (C) a hardware-supported solution involving busy waiting (D) interrupt disabling only

SECTION 9 — DATABASES

- Q185.** Which statement about TRUNCATE is correct?
(A) It is a DDL command that removes all rows of a table (B) It drops the table structure as well (C) It is a DML command that can be rolled back easily (D) It deletes selected rows using a WHERE clause
- Q186.** A view in SQL is
(A) a backup file (B) a virtual table derived from a query (C) an index (D) a physically stored table
- Q187.** Referential integrity in a relational database is enforced through
(A) a check constraint (B) a unique constraint (C) a foreign key (D) a primary key
- Q188.** R(A, B, C, D) has A as its only candidate key. The number of superkeys of R is
(A) 4 (B) 16 (C) 8 (D) 15
- Q189.** Which of the following is NOT a primitive (basic) operation of relational algebra?
(A) Projection (B) Union (C) Selection (D) Natural join
- Q190.** A decomposition of R into R1 and R2 is lossless if
(A) both are in BCNF (B) $R1 \cap R2$ is empty (C) the common attributes form a superkey of at least one of R1 or R2 (D) $R1 \cup R2 \neq R$
- Q191.** A schedule is said to be serializable if it is equivalent to
(A) a schedule with no read operations (B) any schedule without aborts (C) a concurrent schedule with locks (D) some serial schedule of the same transactions
- Q192.** The minimum possible number of tuples in the natural join of two non-empty relations is
(A) 0 (B) the product of the two sizes (C) the size of the smaller relation (D) 1

SECTION 10 — COMPUTER NETWORKS

- Q193.** The well-known default port number for HTTP is
(A) 80 (B) 21 (C) 443 (D) 25
- Q194.** Distance vector routing suffers from which problem?
(A) Hidden terminal (B) Count-to-infinity (C) Silly window syndrome (D) Flooding storm
- Q195.** The TTL field in an IPv4 header prevents
(A) congestion (B) packets from circulating indefinitely (C) fragmentation (D) duplicate acknowledgements
- Q196.** According to the Nyquist criterion, the maximum data rate of a noiseless channel of bandwidth H with V discrete signal levels is
(A) $H \log_2 V$ (B) $2H \log_2 V$ (C) $H \log_2 (1 + S/N)$ (D) $2H V$
- Q197.** The well-known port number used by SMTP is
(A) 25 (B) 23 (C) 21 (D) 110
- Q198.** In Go-Back-N ARQ using 3-bit sequence numbers, the maximum sender window size is
(A) 3 (B) 4 (C) 8 (D) 7
- Q199.** Network Address Translation (NAT) operates at which layer?
(A) Data link (B) Application (C) Network (D) Session
- Q200.** The Address Resolution Protocol (ARP) maps
(A) an IP address to a MAC address (B) a domain name to an IP address (C) a port number to a process (D) a MAC address to an IP address

ANSWER KEY — MOCK CBT 3

Paper-I (Q1 - Q100)

1-B 2-A 3-C 4-B 5-C 6-D 7-D 8-D 9-D 10-B
11-B 12-A 13-C 14-D 15-D 16-D 17-B 18-C 19-A 20-D
21-D 22-A 23-C 24-A 25-D 26-D 27-D 28-A 29-B 30-A
31-D 32-C 33-B 34-D 35-B 36-C 37-B 38-D 39-A 40-A
41-C 42-A 43-D 44-A 45-C 46-B 47-A 48-C 49-C 50-A
51-C 52-C 53-D 54-A 55-A 56-D 57-D 58-A 59-A 60-B
61-D 62-B 63-D 64-B 65-D 66-C 67-D 68-C 69-A 70-A
71-B 72-B 73-D 74-C 75-D 76-D 77-D 78-A 79-A 80-D
81-B 82-D 83-D 84-A 85-A 86-C 87-C 88-A 89-D 90-A
91-D 92-A 93-C 94-C 95-A 96-D 97-A 98-D 99-D 100-A

Paper-II (Q101 - Q200)

101-D 102-A 103-A 104-B 105-D 106-D 107-A 108-A 109-D 110-A
111-D 112-D 113-A 114-C 115-A 116-C 117-A 118-A 119-B 120-A
121-C 122-C 123-C 124-D 125-D 126-B 127-B 128-B 129-A 130-A
131-B 132-C 133-C 134-B 135-D 136-A 137-A 138-A 139-C 140-D
141-A 142-D 143-B 144-B 145-D 146-A 147-D 148-C 149-C 150-A
151-C 152-D 153-B 154-B 155-A 156-A 157-C 158-A 159-C 160-B
161-A 162-C 163-C 164-D 165-B 166-C 167-B 168-D 169-D 170-B
171-A 172-C 173-A 174-B 175-B 176-D 177-D 178-B 179-D 180-A
181-D 182-A 183-D 184-C 185-A 186-B 187-C 188-C 189-D 190-C
191-D 192-A 193-A 194-B 195-B 196-B 197-A 198-D 199-C 200-A

Score sheet

Paper-I: ____ / 100 (qualifying: 40 UR/EWS, 35 OBC-NCL, 30 SC/ST/PwD)

Paper-II: ____ / 100 (same qualifying marks)

Total: ____ / 200

Section-wise wrong answers — GA ____ Reasoning ____ Numerical ____ English ____ | Engg Maths ____ Digital Logic ____ COA ____ Prog & DS ____
Algorithms ____ TOC ____ Compilers ____ OS ____ DBMS ____ CN ____